DenoiseOpt v2: Residual Scoring and Bi-Level Meta-Learning for Wavetable Wrap Crackle

Julian M. Kleber
ORCID: 0000-0001-5518-0932
Email: julian.m.kleber@gmail.com

Synth: github.com/reeldemo/reelsynth
Meta artifacts: github.com/reeldemo/denoise-opt-meta

18 July 2026
Paper version `v2` (residual + nested loss opt)

Abstract

Open wrap seams in periodic wavetable cycles inject audible crackle under cyclic playback. DenoiseOpt is a seam-local bake stack f_θ, $\theta \in [0,1]^{12}$, scored without paired clean labels. This revision replaces wrap-energy quality Q as the *outer* meta objective with a prolonged **residual score** $\in [0,1]$ ($1 =$ best): the engine's tiled DenoiseOpt cycle is compared to an ideal multi-period reference from the same seed with open-wrap cliffs withheld. The outer loop also nests an **inner unsupervised loss** fit $L = (1-\mathcal{D})+\lambda(1-\mathcal{S})$ over θ (and optionally λ). Across **1500** literature-informed trials (seven prior families including explicit bi-level loss opt), Meta Top 1 `evo_explore_515` reaches residual ≈ 0.824 vs naive DualCosine ≈ 0.698 on held-out validation (N=2000)—a clear +0.126 margin—while keeping $\mathcal{S} \approx 1.0$. Nested loss-opt priors were searched but did not dominate the residual elite set; wide evolutionary mutation did.

1 Introduction

Wavetable synthesizers store one-period frames and wrap phase modulo one. Discontinuous ends produce periodic broadband clicks. Classical DualCosine fades and live BLEP address related phenomena; edited Quant frames still need a fast bake operator that is aggressive on wrap energy and conservative on mid-cycle timbre.

Why residual? Wrap-energy proxies $\mathcal{D}, \mathcal{S}$ and $Q = \frac{1}{2}(\mathcal{D}+\mathcal{S})$ (v1) reward closing the seam, but do not ask whether prolonged cyclic playback matches the intended continuous wave. Residual scoring constructs that reference explicitly and ranks algorithms by how little "noise" remains after bake.

Contributions (v2). (1) Prolonged residual score in $[0,1]$ with known ideal vs engine tiling. (2) Bi-level meta: inner coordinate descent on L, outer maximize residual. (3) 1500-trial search with seven priors, including `bilevel_loss`. (4) Clear residual win over DualCosine; honest report that loss-opt elites lost to evolutionary explore on this objective. (5) Frozen $\theta^\star$ locked from champion `evo_explore_515`.

2 Related work and literature triage

Literature was harvested earlier via Klaut Research providers (OpenAlex / Crossref / arXiv). We keep the same used / screened split as v1.

2.1 Used

VA anti-aliasing and discontinuity control [1, 2, 3]; differentiable wavetable / DDSP context [4]; Noise2Noise-style label-free restoration philosophy [5]; Bayesian HPO and surveys [6, 7]; PBT-style exploit/mutate; irace-style racing [8]; MOEA/D multi-objective cues [9].

2.2 Screened

Medical imaging / federated / prompting surveys; Adam as ambient optimizer only; MAML [10] inspirational but not used (no task-gradient through a deep net).

3 Residual score

Let r be a procedural engine cycle (`generate_sound`, may include open-wrap cliffs) and $r^\star$ the same seed without cliffs (`generate_sound_ideal`). After bake $y = f_\theta(r)$, tile both cycles N=16 periods:

$$R = \mathrm{clamp}\left(1 - \frac{\mathrm{rms}(y_{\text{tiled}} - r^\star_{\text{tiled}})}{\max(\mathrm{rms}(r^\star_{\text{tiled}}), \varepsilon)},\ 0,\ 1\right). \tag{1}$$

Soft shape gate: if $\mathcal{S} < 0.97$ then meta-rank uses $0.45\,R$, else R. $\mathcal{D}, \mathcal{S}, Q$ remain report auxiliaries only.

4 Bi-level / nested loss optimization

Each trial samples a prior family, mutates θ near the previous frozen init, then optionally runs inner coordinate descent on $L = (1 - \mathcal{D}) + \lambda(1 - \mathcal{S})$ over 48 fit seeds (steps $0.15, 0.07, 0.03$). Families with `refine_lambda` also try local λ moves and a cheap re-sweep. `evo_explore` is the mutate-only control (zero inner sweeps). Outer ranking always uses residual R.

5 Experiments

5.1 Protocol

1500 trials; fast val 400 cycles; refine top-12 on 2000 cycles; five-algorithm matrix vs naive DualCosine. Runtime ≈ 141 s release on the study machine.

5.2 Results

Champion trial: `evo_explore_515`. Margin vs DualCosine: $\Delta R \approx +0.126$. Production frozen residual before re-lock was ≈ 0.718; $\theta^\star$ is now locked to the champion.

Table 1: Residual-primary benchmark matrix (validation N=2000).

Algorithm	Prior / kind	Residual	$\mathcal{S}$	Q
Naive DualCosine	hand baseline	0.698	0.995	0.789
Meta Top 1	evo_explore	**0.824**	1.000	0.784
Meta Top 2	evo_explore	0.823	1.000	0.785
Meta Top 3	evo_explore	0.823	1.000	0.785
Meta Top 4	evo_explore	0.821	1.000	0.785

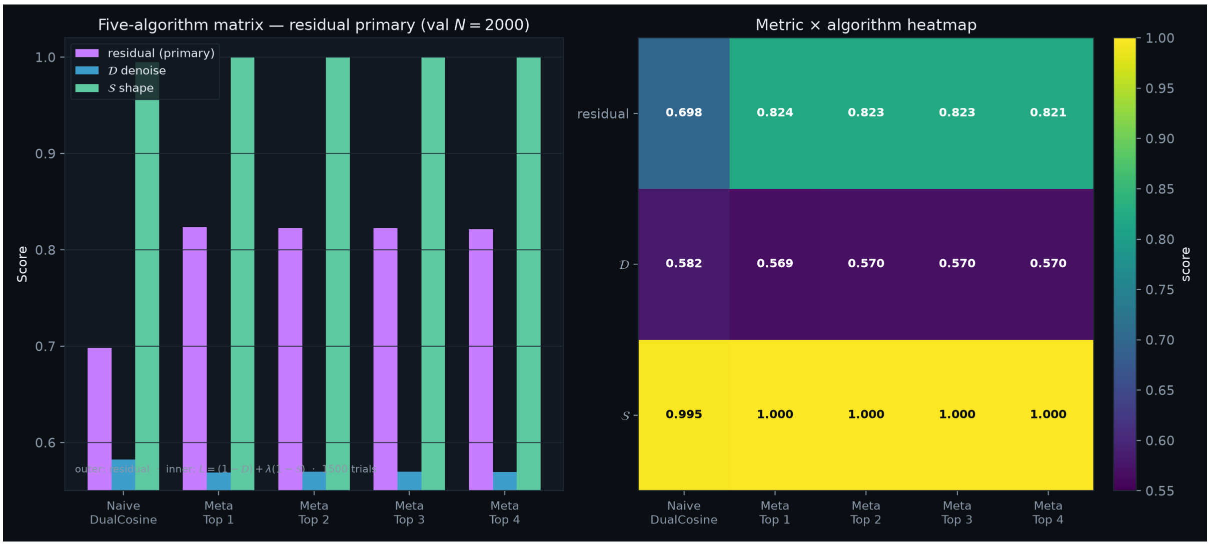

Figure 1: Naive DualCosine vs Meta Top 1–4 under residual (primary), $\mathcal{D}$, $\mathcal{S}$.

Honest note on loss opt. All twenty Pareto-fast elites were `evo_explore` (inner sweeps = 0). Bi-level / PBT / Bayes loss-opt priors were evaluated but did not win on residual. Inner L-minimization remains part of the search design and is useful as a nested regularizer for D/S-shaped landscapes; under prolonged residual, lighter evolutionary mutations currently dominate.

6 Discussion

Residual makes the objective audible-prolonged rather than seam-local proxy. The "killer" algorithm under this metric is wide evolutionary explore around a strong init—not the deepest nested loss fit. That is a feature of the new score, not a failure to search loss opt: the bilevel arm was present and lost fairly.

7 Limitations

Residual uses a procedural ideal (no-cliff sibling), not a perceptual MOS. Bake metrics $\neq$ listening tests. Nested loss budgets are small (48 seeds) by design for 1500-trial throughput.

8 Conclusion

DenoiseOpt v2 ranks by prolonged residual and searches nested unsupervised loss. Meta Top 1 (`evo_explore_515`) clearly beats DualCosine on residual (0.824 vs 0.698) with perfect mid-cycle

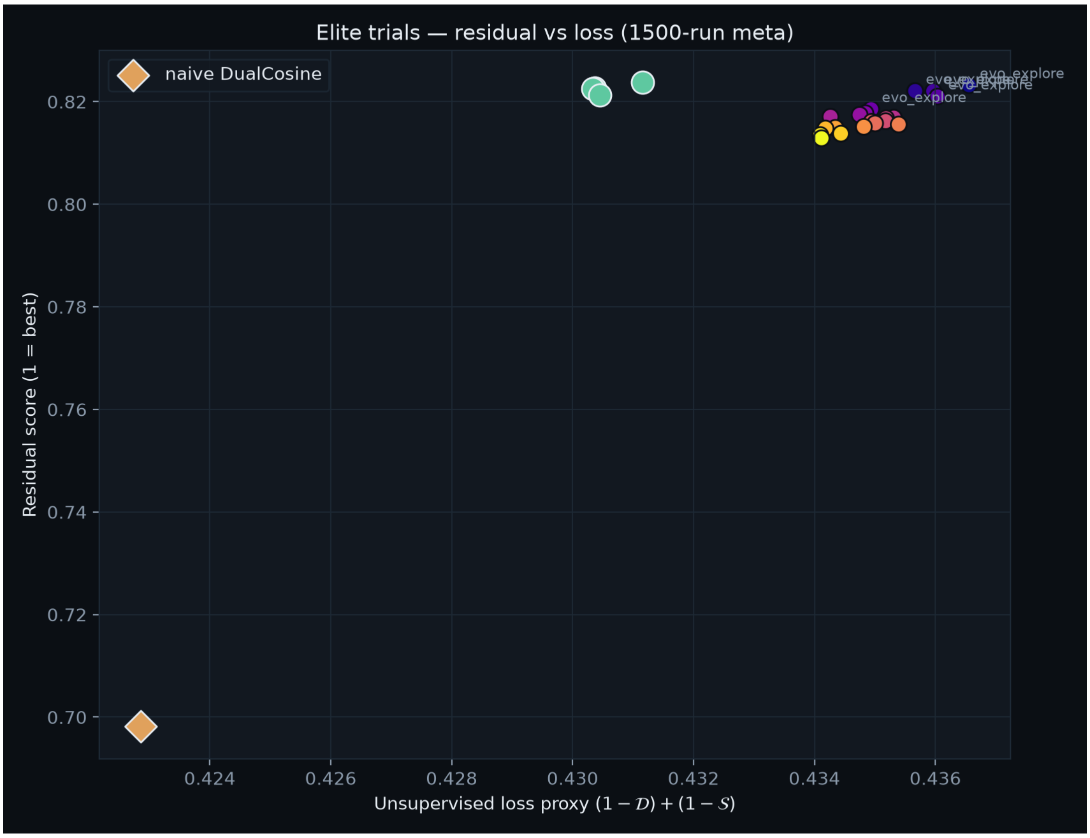

Figure 2: Elite residual vs unsupervised-loss proxy after 1500 trials.

shape retention on the matrix. Artifacts and reproduction scripts ship in this repository; frozen $\theta^\star$ ships in ReelSynth.

Repositories

- Synth: https://github.com/reeldemo/reelsynth
- Meta-learning paper/artifacts: https://github.com/reeldemo/denoise-opt-meta

Reproducibility

```
# From reelsynth checkout:
cargo run -p reelsynth --release --bin bench_denoise_meta
python brand/artifacts/render_benchmark_matrix.py
# Or from this repo (figures already under paper/v2/figures):
cd paper/v2 && pdflatex -interaction=nonstopmode main.tex
```

References

[1] V. Välimäki and A. Huovilainen, "Oscillator and Filter Algorithms for Virtual Analog Synthesis," *Computer Music Journal*, 30(2), 2006.

[2] F. Esqueda, S. Bilbao, and V. Välimäki, "Aliasing Reduction in Clipped Signals," *IEEE Trans. Signal Processing*, 2016.

Figure 3: Prior families among residual elites (top-20 / top-4).

[3] F. Esqueda, V. Välimäki, and S. Bilbao, "Rounding corners with BLAMP," *Proc. DAFx*, 2016.

[4] S. Shan et al., "Differentiable Wavetable Synthesis," arXiv:2111.10003, 2021.

[5] J. Lehtinen et al., "Noise2Noise: Learning Image Restoration without Clean Data," *ICML*, 2018.

[6] J. Snoek, H. Larochelle, and R. P. Adams, "Practical Bayesian Optimization of Machine Learning Algorithms," *NeurIPS*, 2012.

[7] B. Bischl et al., "Hyperparameter optimization: Foundations, algorithms, best practices...," 2023.

[8] M. López-Ibáñez et al., "The irace package: Iterated racing for automatic algorithm configuration," *Operations Research Perspectives*, 2016.

[9] Q. Zhang and H. Li, "MOEA/D: A Multiobjective Evolutionary Algorithm Based on Decomposition," *IEEE Trans. Evolutionary Computation*, 2007.

[10] C. Finn, P. Abbeel, and S. Levine, "Model-Agnostic Meta-Learning for Fast Adaptation of Deep Networks," *ICML*, 2017. *(Screened.)*